Md Mursalatul Islam Pallob

+8801611-920058 | mursalatul.pallob@gmail.com | linkedin.com/in/mursalatul | github.com/mursalatul |
mursalatul.vercel.app

Education

Dhaka International University — Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering (CGPA 3.60/4.00) — *Aug. 2021 – Oct 2025*

Experience

Brace Laboratory (bracelaboratory.com) — Dec 2024 – Present
Founder & Chief Executive Officer — *Dhaka, Bangladesh*

- Spearhead company operations, strategic vision, and technical architecture for software development and active research projects.
- Oversee cross-functional team workflows, ensuring high-quality product delivery while maintaining an internal policy supporting active research initiatives.

Medvisor Mobile — Jun 2024 – Aug 2024
Lead Software Engineer intern (Part-time) — *Irvine, CA*

- Developed and implemented machine learning models for voice and emotion stress analysis.
- Collaborated closely with Flutter, DevOps, and UX/UI engineers to integrate AI-driven features into mobile apps, aligning with project goals.

Projects

Suicidal Ideation Prediction | *Python, Natural Language Processing, Machine Learning* — Aug 2025 – Oct 2025

- Built an end-to-end NLP pipeline in Python for preprocessing text data, tokenization, and feature extraction to support supervised classification of suicidal ideation from short-form text.
- Implemented and compared modeling approaches (e.g., transformer-based fine-tuning and classical ML baselines) and established a reproducible training/evaluation workflow using cross-validation and standard metrics (precision, recall, F1, AUC).
- Packaged experiments and analysis in reproducible notebooks and scripts, and documented data handling and ethical/privacy considerations to support responsible use and further research.

Infant Cry Analysis Web Application | *Python, Django, PostgreSQL, Bootstrap* — May 2025 – Oct 2025

- Developed a full-stack web application (using Django and PostgreSQL) for the detection and multi-class diagnosis of infant distress from cry recordings, providing a user-friendly interface for uploading and analyzing audio data.
- Integrated a machine learning backend capable of processing audio features (like Spectro-Derivative features) to predict and classify cries into different distress categories (e.g., pain, hunger, sleepy, healthy).
- Designed a responsive frontend using Bootstrap to display classification results, visualizations of audio features, and model performance metrics.

Infant Cry Analysis ML Models | *Python, Scikit-learn, Spectrograms, Machine Learning* — Mar 2025 – Oct 2025

- Designed and implemented a Balanced Augmented Spectro-Derivative Ensemble Framework for multi-class infant distress diagnosis, achieving high classification accuracy.
- Extracted and analyzed spectrograms and Spectro-Derivative features from infant cry audio for effective feature engineering and representation.
- Developed and evaluated various machine learning models (e.g., ensemble methods) for the classification of infant cries into specific distress types, focusing on robust performance across different datasets.

Real-time Speech Emotion Recognition | *Python, Keras, LSTM, Audio Processing* — Jun 2024 – Jul 2024

- Developed a real-time Speech Emotion Recognition system using an LSTM-based deep learning model implemented in Keras to analyze live microphone input and identify expressed emotions.
- Prepared and augmented audio datasets (RAVDESS, TESS), extracted acoustic features and spectrogram representations, and engineered time-series inputs suitable for recurrent neural networks.
- Delivered a real-time inference pipeline with continuous monitoring, achieved 87% accuracy on benchmark evaluations, and implemented silence detection to automatically stop processing for robustness.

Achievements

- Winner of the Programming Competition, June 2022 — DIU.
- 2nd Place in Programming Competition at IT Carnival 2022 — DIU.
- ICPC 2021 Onsite Contestant (National Rank: 90) — BUBT.
- SUST IUPC 2023 (National Rank: 82) — SUST.
- Qualified for ICPC 2022 Onsite Contest — Green University of Bangladesh.
- NASA International Space Apps Challenge 2023 — Global Nominee; 1st Runner Up, Barishal Division, Bangladesh.
- Qualified for ICPC 2023 Onsite Contest — BUBT.
- NASA International Space Apps Challenge 2024 — 2nd Runner Up, Barishal Division, Bangladesh.
- KUET IUPC 2025 — Participated in Inter-University Programming Contest.
- UIU IUPC 2025 (National Rank: 60) — United International University.
- NASA International Space Apps Challenge 2025 — Honorable Mention; Global Nominee; Champion, Barishal Division, Bangladesh.

Leadership & Activities

DIU Computer Programming Club — Dhaka, Bangladesh

Vice President — Sep 2024 – Mar 2025

- Oversaw club activities, including organizing coding competitions, workshops, and hackathons, to foster collaboration and skill development.
- Collaborated with the executive team to create opportunities for members to advance their programming knowledge and careers, building a strong community.
- Organized DIU CSE Fall Fest 2024.

Competition Lead — Oct 2023 – Aug 2024

- Managed diverse programming contests and authored 59 original competitive programming problems for club competitions at Dhaka International University.
- Mentored club members in competitive programming, providing structured training to prepare teams for prestigious competitions including ICPC and IUPC.

IEEE DIU Student Branch — Dhaka, Bangladesh

Secretary General — Sep 2021 – Sep 2023

- Organized seminars and fundraising events to facilitate student branch initiatives.

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript

Machine Learning & AI: Classical ML, Deep Learning (CNN, LSTM, Transformers), NLP, Audio Signal Processing, Feature Engineering, Model Evaluation

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, Django, Numpy, Pandas

Data & Tools: PostgreSQL, Git, GitHub, Docker, VS Code, PyCharm

Languages

- English - Professional working proficiency
- Bangla - Native

Publications

[1] M. M. I. Pallob, F. B. Orpita, N. Usha, N. Hasan, M. N. Tahira, and M. A. Based, "Infant Cry Analysis Using Machine Learning Models and Extended Feature Selection," in *Proceedings of BIM 2025*, Taylor & Francis, 2025.

[2] A. M. Rion, M. M. I. Pallob, R. Rakib, M. A. Molla, M. Hasan, and M. A. Based, "Cross-Cultural Early Detection of Suicidal Ideation in Adolescents: An Ensemble Machine Learning Approach Using GSHS Data," in *Proc. IEEE TENCON*, Bali, Indonesia, 2026.

[3] M. A. Molla, R. Rakib, M. M. I. Pallob, M. Hasan, A. M. Rion, and M. A. Based, "FairCF: Fair Counterfactual Explanations for Student Academic Performance Prediction," in *Proc. IEEE ICAIMS*, 2026.

[4] M. M. I. Pallob, M. A. Molla, R. Rakib, and M. A. Based, "AI-Driven Prescriptive Analytics for Hybrid Learning Mode Optimization: A Multi-Model Benchmarking and Explainable AI Framework," in *Proc. IEEE ICAIMS*, 2026.

[5] M. A. Molla, R. Rakib, M. Hasan, M. M. I. Pallob, A. M. Rion, and M. A. Based, "A Hybrid Explainable AI Framework for Concurrent Earthquake Magnitude Estimation and Severity Classification," in *Lecture Notes in Networks and Systems* (MIET 2026), vol. XXX, Cham, Switzerland: Springer, 2026.

— I hereby declare that all information presented in this resumé is accurate, complete, and a truthful reflection of my qualifications, experiences, and achievements. —

— Md Mursalatul Islam Pallob —